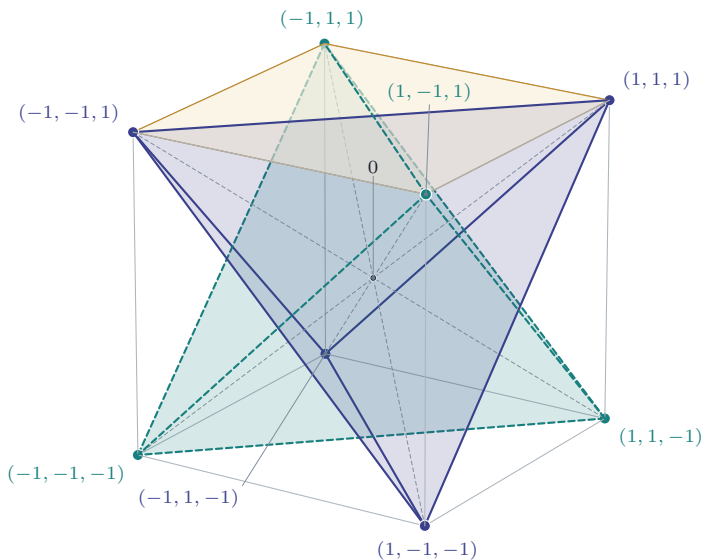




At a CM point c

vertices: the eight weights of the Hodge torus;
 $\text{End}^0(X_c) \otimes \mathbb{Q}$ is the algebra of functions
 on them, diagonal in the weight basis

diagonals: the opposite pairs $\{w, -w\}$,
 $\psi(e_w, e_{-w}) = \pm 1$; functions on the four
 diagonals: $K_c^+ = \text{End}^0(X_c)^\dagger$, totally real,
 of degree four

tetrahedra: T_+ (solid), $T_- = -T_+$ (dashed),
 the zero-sum four-sets with no opposite pair;
 functions on the two tetrahedra: k_c ,
 imaginary quadratic

$t_\pm = \prod_{w \in T_\pm} e_w$ span the Weil line $W(X_c)$

shaded face: $H^{1,0}(X_c) = V_1 \otimes V_2 \otimes V_3^{1,0}$,
 two vertices in each tetrahedron

$\text{End}^0(X_c) = K_c^+ \otimes k_c$: a vertex is
 its diagonal together with its tetrahedron

the Galois group acts linearly, permuting the
 vertices; it preserves the diagonals and $\{T_+, T_-\}$